

Matthew Maloni

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EDUCATION

University of Florida, College of Liberal Arts and Sciences

August 2023 – December 2026

Bachelor of Science in Data Science

Gainesville, FL

GPA: 3.62 / 4.00 · Dean's List: Fall 2023, Fall 2025

Relevant Coursework: Applied Machine Learning Systems (EEL5934, graduate section); Data Structures & Algorithms; Linear Algebra for Data Science; Probability; Linear Regression; Programming with Data in R

WORK EXPERIENCE

Kore Capital

May 2026 – Present

Financial Engineering Intern

Boynton Beach, FL

- Co-developed a linear-regression equity factor model spanning momentum, value, quality, size, and growth baskets, applying R^2 and autocorrelation testing on Bloomberg data to evaluate hedge quality and factor attribution.
- Integrated Claude MCP servers into the firm's .NET portfolio system and Excel add-in, condensing tax-lot, PnL, and reporting workflows into automated tools that surface substrategy-level analytics.
- Consolidated 100K+ position, tax-lot, and trade records into a unified analytics layer, building per-campaign visualizations that expose trading patterns and adverse-move risk.

Pastry Art Wine Club

October 2025 – Present

Software Developer

Sarasota, FL (Remote)

- Built and shipped a production iOS app in React Native and Expo for a wine retailer's subscription club, owning it end to end through App Store submission and release.
- Integrated the OpenAI API for wine-label photo analysis and a personalized recommendation chatbot, backed by fuzzy matching against the store's catalog on a Supabase/PostgreSQL backend.

Trustwell

April 2025 – August 2025

Data Science Intern

Salem, OR (Remote)

- Compiled multi-region regulatory datasets and integrated the OpenAI API to improve response quality for Reggie, Trustwell's food-safety compliance assistant; supported the AngularJS-to-Vue-3 migration.

RESEARCH & LEADERSHIP

IEEE Signal Processing Society, UF Student Chapter

September 2025 – Present

Research Board, Project Lead

Gainesville, FL

- Founded and lead a five-member research group building binaural (dual-microphone) audio scene analysis for real-world sound-source localization and separation.
- Implemented scoring and trust-gating algorithms for Plato's Cave, a knowledge-graph paper-verification system; co-authored the resulting paper, accepted to the AAAI Spring Symposium (AAAI-MAKE) 2026.

Nettles Lab, UF Scripps Institute

January 2026 – May 2026

Undergraduate Researcher

Jupiter, FL

- Trained a VAMPnet in PyTorch mapping molecular dynamics trajectories to metastable conformational states for NIH-funded glucocorticoid receptor drug discovery, reaching a VAMP-2 score of 2.87 against a max of 3.0.

Beta Theta Pi, Gamma Xi Chapter

August 2025 – Present

Internal Philanthropy Chair

Gainesville, FL

- Directed Beta Bring It On 2026, raising a record \$64,000 for the Alzheimer's Association and selling out the Phillips Center's 1,700 seats with 300+ participants across 19 teams.

PROJECTS

CNN Medical Image Classification

November 2025 · TensorFlow/Keras, HiPerGator

- Compared four CNN architectures on chest X-ray and diabetic retinopathy classification, achieving 94.77% accuracy and 83.61% AUC across 14 thoracic pathologies over 78,468 images.

Dimensionality Reduction for Satellite Ship Detection

November 2025 · Python, scikit-learn, HiPerGator

- Benchmarked nine PCA/ISOMAP and classifier configurations on 4,000 satellite images, achieving 96.75% accuracy at $82\times$ baseline throughput with PCA and SVM, while the top-accuracy model detected only 1–2 ships per scene in practice.

SKILLS

Languages: Python, C++, R, SQL, JavaScript, MATLAB, Bash

Machine Learning: PyTorch, TensorFlow/Keras, scikit-learn, CNNs, self-supervised learning

Infrastructure & AI Systems: AWS, PostgreSQL, Supabase, Git/GitHub, Docker, Bloomberg Terminal, OpenAI API, Claude/MCP

Web & Mobile: React Native, Expo, React, Vue 3, Node.js